

Supplemental Material: Spatial Scrambling Profiles and System-Size Scaling of Ω across Dynamical Regimes

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(Dated: March 2026)

This Supplemental Material contains: (S1) complete $\Omega_{\text{late}}(r)$ profiles for all five models and all system sizes; (S2) full cross-platform hardware data tables; (S3) fit diagnostics and AIC computation; (S4) circuit and simulation parameters.

All numerical values in this document are extracted programmatically from the authoritative data files (`paper1b_master_5models.json`, `aqt_results.json`, `rigetti_results.json`) and verified against independent recomputation. The extraction script (`generate_figures.py`) is included in the repository.

COMPLETE SPATIAL PROFILES $\Omega_{\text{late}}(r)$

Tables I–V give the complete $\Omega_{\text{late}}(r)$ profiles for each model. All values from exact statevector simulation with $D_{\text{max}} = 5N$, dense trace method ($N \leq 12$) or typicality with 10 random states ($N \geq 14$, KI chaotic only).

TABLE I. KI chaotic $\Omega_{\text{late}}(r)$ profiles, $r = 1$ –7. Dash (—) indicates $r \geq N$.

N	$r=1$	$r=2$	$r=3$	$r=4$	$r=5$	$r=6$	$r=7$
4	0.280	0.326	0.331	—	—	—	—
6	0.215	0.252	0.274	0.275	0.402	—	—
8	0.159	0.191	0.210	0.222	0.244	0.262	0.393
10	0.135	0.162	0.180	0.186	0.197	0.210	0.227
12	0.119	0.140	0.156	0.159	0.168	0.176	0.185

TABLE II. KI chaotic $\Omega_{\text{late}}(r)$ profiles, $r = 8$ –11. Only $N = 10$ and $N = 12$ have data in this range.

N	$r=8$	$r=9$	$r=10$	$r=11$
10	0.247	0.376	—	—
12	0.196	0.212	0.234	0.360

Note: the last distance ($r = N - 1$) in OBC consistently shows a jump in Ω_{late} (e.g., $N = 6$: $r = 4$ gives 0.275 but $r = 5$ gives 0.402; $N = 12$: $r = 10$ gives 0.234 but $r = 11$ gives 0.360). This boundary effect, where the far-end operator is partially shielded from scrambling by its proximity to the open boundary, is present at all system sizes and should not be confused with the monotonic increase due to butterfly velocity at intermediate distances.

Integrable and dual-unitary models give $\Omega_{\text{late}}(r) = 1.000$ to machine precision ($< 10^{-13}$) for all N and r ; these values are omitted for brevity.

CROSS-PLATFORM HARDWARE DATA

Tables VI and VII give the full $\langle Z_0 \rangle$ measurements for $N = 4$ across all circuit types and depths. Each row cor-

TABLE III. KI mixing $\Omega_{\text{late}}(r)$ profiles, $r = 1$ –7. The U-shaped pattern (minimum at interior, maxima at boundaries) persists across all N but becomes asymmetric for $N \geq 10$: the far-boundary values are lower than the near-boundary values.

N	$r=1$	$r=2$	$r=3$	$r=4$	$r=5$	$r=6$	$r=7$
4	0.533	0.461	0.820	—	—	—	—
6	0.865	0.599	0.472	0.632	0.897	—	—
8	0.953	0.824	0.618	0.536	0.579	0.738	0.916
10	0.977	0.922	0.786	0.622	0.573	0.586	0.660
12	0.985	0.956	0.895	0.764	0.630	0.588	0.593

TABLE IV. KI mixing $\Omega_{\text{late}}(r)$ profiles, $r = 8$ –11. Asymmetry visible: compare $r = 9$ at $N = 10$ (0.932) with $r = 1$ (0.977); compare $r = 11$ at $N = 12$ (0.952) with $r = 1$ (0.985).

N	$r=8$	$r=9$	$r=10$	$r=11$
10	0.811	0.932	—	—
12	0.635	0.725	0.853	0.952

responds to one circuit: the magnetization (mag) circuit measures $\langle Z_0 \rangle$ after forward evolution $U_F^d |\psi_0\rangle$; the echo circuit measures $\langle Z_0 \rangle$ after the full echo $U_F^{-d} Z_r U_F^d |\psi_0\rangle$ with $r = 1$.

The key comparison is the bolded entries: KI chaotic $d = 3$ magnetization has $|\Delta| = 0.015$ (AQT) vs. $|\Delta| = 0.219$ (Rigetti), a $15\times$ improvement. Echo circuits are more demanding (double the effective circuit depth) and both platforms show larger errors for echo at $d \geq 3$.

Note on IBM data. IBM `ibm_marrakesh` data (429 circuits, Conjunto A parameters) are reported in full in Paper I [1]. They are included here only in aggregated form (Table IV of the main text) for protocol comparison. IBM data use Conjunto A; Rigetti and AQT use Conjunto B. The cross-platform comparison validates the measurement protocol, not point-by-point physical equivalence.

Note on echo $q1$ – $q3$ in AQT integrable. For all three depths, AQT IBEX Q1 measures $\langle Z \rangle = 1.000$ exactly on qubits 1, 2, and 3 in the integrable echo circuit. With

TABLE V. SYK $\Omega_{\text{late}}(r)$ profiles. Disorder average over 10 seeds. Flat profile to within $\lesssim 10^{-3}$ at each N , confirming absence of spatial structure from all-to-all connectivity.

N	$r=1$	$r=2$	$r=3$	$r=4$	$r=5$	$r=6$	$r=7$	$r=8$	$r=9$
4	0.072	0.072	0.071	—	—	—	—	—	—
6	0.016	0.018	0.019	0.020	0.019	—	—	—	—
8	0.005	0.005	0.004	0.005	0.005	0.005	0.005	—	—
10	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001

TABLE VI. AQT IBEX Q1: $\langle Z_0 \rangle$ measurements vs. exact simulation. $N = 4$, 100 shots per circuit ($\sigma \approx 0.10$). Values are qubit-0 magnetization.

Model	Type	d	Exact	AQT	$ \Delta $
Integrable	mag	1	+0.540	+0.500	0.040
Integrable	echo	1	-0.416	-0.280	0.136
Integrable	mag	3	-0.990	-1.000	0.010
Integrable	echo	3	+0.960	+0.880	0.080
Integrable	mag	5	+0.284	+0.340	0.056
Integrable	echo	5	-0.839	-0.820	0.019
KI chaotic	mag	1	+0.540	+0.420	0.120
KI chaotic	echo	1	-0.416	-0.440	0.024
KI chaotic	mag	3	+0.575	+0.560	0.015
KI chaotic	echo	3	+0.126	-0.140	0.266
KI chaotic	mag	5	+0.550	+0.420	0.130
KI chaotic	echo	5	+0.074	+0.000	0.074

$J = 0$ (no entangling gates), the perturbation Z_1 at site 1 does not propagate to distant qubits; the echo should return them to the initial state $|0\rangle$ with $\langle Z \rangle = 1$ exactly. That 100 out of 100 shots give $|0\rangle$ on all three qubits at all depths confirms exceptional single-qubit fidelity of the ion trap.

FIT DIAGNOSTICS

AIC Computation

The Akaike Information Criterion is $\text{AIC} = 2k - 2 \ln \hat{L}$, where k is the number of free parameters and $\hat{L}$ is the maximized likelihood. For least-squares fits to n data points with residual sum of squares RSS:

$$\text{AIC} = n \ln(\text{RSS}/n) + 2k. \quad (1)$$

Both algebraic (Hip. A) and power-law (Hip. C) have $k = 2$ free parameters, so ΔAIC depends only on the ratio of RSS values. With $\text{RSS}_A = 3.21 \times 10^{-5}$ and $\text{RSS}_C = 4.67 \times 10^{-5}$ for $n = 8$:

$$\Delta\text{AIC} = 8 \ln\left(\frac{3.21}{4.67}\right) = -3.00. \quad (2)$$

By convention, $|\Delta\text{AIC}| > 2$ constitutes positive evidence and $|\Delta\text{AIC}| > 6$ constitutes strong evidence. The exact RSS values from the numerical fit (before rounding) give

TABLE VII. Rigetti Ankaa-3: $\langle Z_0 \rangle$ measurements vs. exact simulation. $N = 4$, 2048 shots ($\sigma < 0.02$). Values are qubit-0 magnetization.

Model	Type	d	Exact	Rigetti	$ \Delta $
Integrable	mag	1	+0.540	+0.517	0.024
Integrable	echo	1	-0.416	-0.376	0.040
Integrable	mag	3	-0.990	-0.867	0.123
Integrable	echo	3	+0.960	+0.924	0.036
Integrable	mag	5	+0.284	+0.248	0.036
Integrable	echo	5	-0.839	-0.731	0.108
KI chaotic	mag	1	+0.540	+0.546	0.006
KI chaotic	echo	1	-0.416	-0.077	0.339
KI chaotic	mag	3	+0.575	+0.356	0.219
KI chaotic	echo	3	+0.126	-0.125	0.251
KI chaotic	mag	5	+0.550	+0.112	0.437
KI chaotic	echo	5	+0.074	-0.032	0.106

$\Delta\text{AIC} = -3.30$, reported throughout the main text; the rounded values quoted above yield -3.00 as an approximate illustration of the computation. The value -3.30 represents moderate positive evidence in favor of Hip. A.

Exponential Exclusion (Hip. B)

The exponential fit $\Omega = e^{-\gamma N}$ was tested on the same 8-point series. Best-fit $\gamma = 0.053$, $\text{RSS}_B = 2.48 \times 10^{-3}$, giving $R^2 = 0.967$ and $\Delta\text{AIC} = +19.1$ relative to power-law. This provides very strong empirical evidence against the exponential form for KI chaotic. For systems with energy conservation, the Huang-Brandão-Zhang theorem [2] provides additional theoretical support; KI chaotic is Floquet (no energy conservation), so its exclusion rests on empirical grounds alone.

β -Function Finite-Difference Table

TABLE VIII. Effective β -function from 8-point KI chaotic series, central differences. Ω column shows the value at the central point; β_{eff} is $(\Omega_{i+1} - \Omega_{i-1})/(\ln N_{i+1} - \ln N_{i-1})$.

Ω	β_{eff}	$-c\Omega(1-\Omega)$
0.215	-0.175	-0.167
0.159	-0.157	-0.132
0.135	-0.098	-0.116
0.119	-0.096	-0.104
0.103	-0.098	-0.091
0.091	-0.085	-0.082

The data show that β_{eff} is systematically more negative than $-c\Omega(1-\Omega)$ at large Ω (small N), which is expected from finite-size effects in the pre-asymptotic regime. At small Ω (large N), the two converge, consistent with the

algebraic fit becoming more accurate as finite-size corrections diminish.

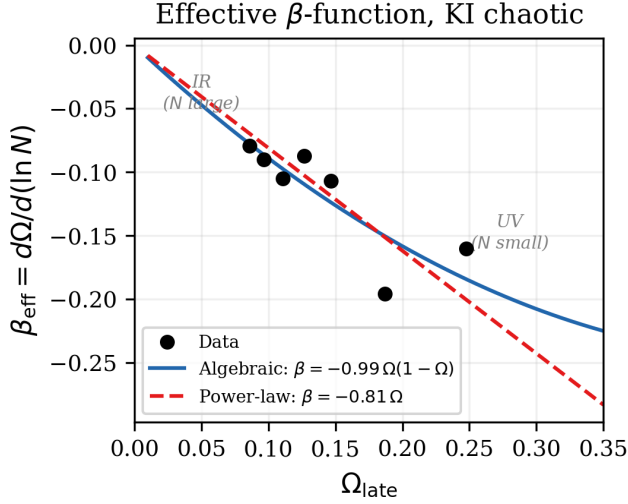


FIG. 1. Effective β -function $\beta_{\text{eff}} = \Delta\Omega/\Delta(\ln N)$ vs. Ω_{late} for KI chaotic (circles). Algebraic fit (solid, $\beta = -c\Omega(1 - \Omega)$, $c = 0.990$) and power-law fit (dashed, $\beta = -\alpha\Omega$, $\alpha = 0.810$) are statistically indistinguishable in the data range $\Omega \in [0.08, 0.28]$.

CIRCUIT AND SIMULATION PARAMETERS

Kicked Ising Circuit (Conjunto B)

The Floquet operator is:

$$U_F = e^{-ih \sum_{i=0}^{N-1} X_i} e^{-ib \sum_{i=0}^{N-1} Y_i} e^{-iJ \sum_{i=0}^{N-2} Z_i Z_{i+1}}, \quad (3)$$

with $J = 1.0$, $h = 0.5$, $b = 0.5$ (KI chaotic), $J = 0.1$ (KI mixing), $J = \pi/4$, $h = \pi/4$, $b = 0$ (dual-unitary), $J = 0$, $b = 0$ (integrable). Open boundary conditions; no $Z_i Z_{i+1}$ term for $i = N - 1$.

Echo Protocol

Initial state: $|\psi_0\rangle = |0\rangle^{\otimes N}$ (all qubits in $|0\rangle$ state). Circuit structure per depth d , scrambled operator at site r :

$$|\phi_r(d)\rangle = U_F^{-d} Z_r U_F^d |\psi_0\rangle. \quad (4)$$

Observable: $\langle Z_0 \rangle = \langle \phi_r(d) | Z_0 | \phi_r(d) \rangle$. Normalization: $C(0) = \langle \psi_0 | Z_0 | \psi_0 \rangle = 1$ (all qubits in $|0\rangle$).

D_{max} and Ω_{late} Computation

For each (N, model) pair, $D_{\text{max}} = 5N$. The late-time average uses depths $d \geq D_{\text{max}}/2$, i.e., the second

half of the circuit depth range. For $N = 4$: depths $d = 10, \dots, 20$ contribute. The dense trace method computes the exact statevector at each depth; the typicality method approximates the trace by averaging over 10 random initial states sampled from the Haar measure on the 2^N -dimensional Hilbert space.

SYK Model

The SYK-inspired Floquet operator uses random all-to-all ZZ couplings: $H_{\text{SYK}} = \sum_{i < j} J_{ij} Z_i Z_j$ with $J_{ij} \sim \mathcal{U}(0.5, 1.5)$ independently for each realization. The Floquet operator is $U_{\text{SYK}} = e^{-iH_{\text{SYK}}}$. Disorder averaging uses 10 independent coupling realizations for all $N \leq 12$.

Hardware Circuit Transpilation

All circuits were transpiled using Qiskit 2.3.0 (IBM), PyQuil (Rigetti via AWS Braket), and the AQT SDK (AQT via AWS Braket), with optimization level 3. The echo protocol was verified with integrable circuits on all platforms before running chaotic models.

REPOSITORY CONTENTS

The data repository [1] contains:

Data files:

- `paper1b_master_5models.json` — authoritative dataset, 5 models, $N = 4-12$, $D_{\text{max}} = 5N$, dense trace
- `ki_chaotic_N14.18.json` — KI chaotic typicality results, $N = 14, 16, 18$
- `syk_n12.results.json` — SYK $N = 12$, 10 seeds, $D_{\text{max}} = 60$
- `aqt_results.json` — 12 AQT circuits, raw magnetizations (all 4 qubits)
- `aqt_analysis_final.json` — 48 AQT measurements with exact-simulation diffs
- `rigetti_results.json` — 26 Rigetti circuits, raw
- `rigetti_analysis_final.json` — Rigetti analysis

Scripts:

- `omega_simulation.py` — exact simulation (dense trace and typicality)
- `scaling_fits.py` — algebraic/power-law fits, ΔAIC , β_{eff} computation

- `cross_platform_analysis.py` — hardware comparison
- `generate_figures.py` — reproduces all figures from JSON files

Documents:

- `paper1b_main.tex` — main manuscript
- `paper1b_supplemental.tex` — this document
- `kaelion_v8.bib` — bibliography (v8, 58 entries)

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- [1] E. F. Perez Eugenio, Probing quantum scrambling via OTOCs in digital circuits: Finite-size recurrences, system-size scaling, and validation on IBM quantum hardware (2026), preprint. arXiv submission in preparation.
- [2] Y. Huang, F. G. S. L. Brandão, and Y.-L. Zhang, Finite-size scaling of out-of-time-ordered correlators at late times, Phys. Rev. Lett. **123**, 010601 (2019).